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PALAEOMAGNETIC STUDIES OF THE KAPITI PHONOLITE OF KENYA

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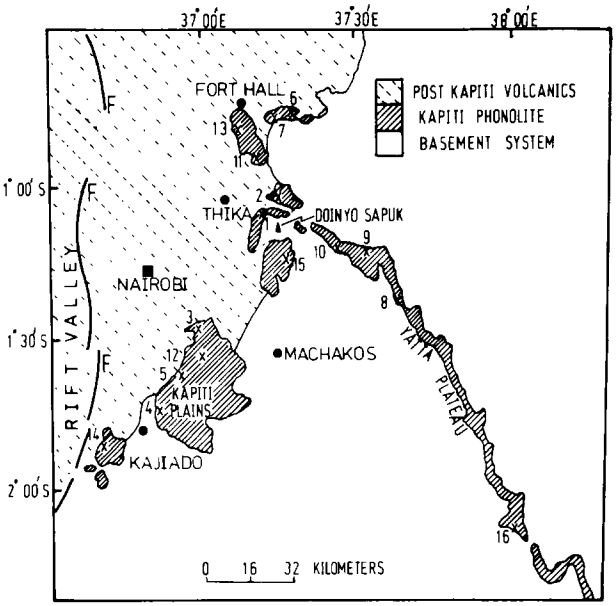
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Specimens of Kapiti phonolite from sixteen sites were treated in an alternating field up to 900 Oe peak. Four sites were discarded on the basis of Watson's Criterion for randomness. Variations in various magnetic parameters along a vertical section of the rock indicate that a hard secondary component of magnetization of high coercive force is present in the surface samples of the rock – a possible cause for the poor grouping of directions of the remaining sites. Weathering may have introduced this component in the surface samples.

The Kapiti phonolite possesses both normal and reverse polarities. The Pole position for the Upper Miocene is calculated at 81°N, 118°E with a circle of confidence of 17°.

1. Introduction

The Kapiti phonolite is one of the earliest Tertiary lavas to be extruded on the eastern flank of the Gregory Rift Valley and rests on the sub-Miocene peneplain. It occurs in three distinct areas: the Kapiti Plains, the Yatta Plateau and the Fort Hall area (Map 1). The surface of the Kapiti Plains is flat with poor exposures of the phonolite. The Yatta Plateau phonolite forms a long narrow ridge (4.8 km) for a distance of 228 km towards south-south east. The ridge probably represents an ancient valley cut in the sub-Miocene surface which was protected from subsequent erosion by infilling with the phonolite [1, 2].



Map 1. Distribution of sampling sites.

The distinctive phonolite is a hard, black, fine-grained rock which consists of very large phenocrysts of anorthoclase and waxy nepheline in a microcrystalline groundmass. A few vesicles are filled by calcite and zeolite.

In a progress report on the Tertiary volcanics in East Africa [3] two polarities of the Kapiti phonolite are quoted. The Yatta Plateau and the Kapiti Plains phonolites have been dated at 13.2 my and 13.4 my by K/Ar in feldspar and nepheline respectively [4]. Bishop and others [5] redated the Kapiti Plains phonolite at 12.9 ± 0.7 my and 13.1 ± 0.5 my using the same method.

2. Sampling

Two hundred and twenty four samples of the Kapiti phonolite were collected from sixteen sites as shown in Map 1. These were collected either as oriented blocks, or as 2.2 cm diameter cores about 9.0 cm long using a portable drill of the type described by Doell and Cox [6].

Freshly exposed rock at roads and railway cuttings, tunnels and quarries was sampled wherever possible. Such sites were difficult to find in this rock unit. Sites 6, 11 and 12 were the only sites from which freshly exposed samples were obtained. Sites 1–4 and 13 were exposures on river banks, and sites 5, 8, 9, 10, 14 and 16 were exfoliated surface outcrops. Outcrops with obvious surface weathering were avoided for sampling.

3. Measurements

The directions of remanent magnetization of several 2.5 cm long specimens from each sample were measured using an astatic magnetometer.

The specimens were "cleaned" magnetically using the alternating field demagnetization technique. The thermal method was avoided due to the explosive nature of the rock. The method employed for the alternating field demagnetization was similar to that used by McElhinny and Gough [7]. Two to three "pilot" specimens from each site were progressively demagnetized using alternating fields. The field at which the directions of remanent magnetization were observed constant or suffering very little change during demagnetization was defined as the "optimum field" for that particular site. All the other specimens from the site were then demagnetized at this field. The optimum field strength for the sites varied from 200 to 500 Oe peak. Curie points were measured on a Curie balance similar to that described by Larochelle [8] and constructed by our colleague Dr. J. Vise. The d.c. field in which these measurements were made was about 1500 Oe.

4. Results and discussion

The directions of natural remanent magnetization remained scattered after cleaning and in four of the sixteen sites were random by the Watson test with a probability of 5% [9]. These four sites, 2, 7, 4 and 9 (table 1), were excluded from the final analysis although the demagnetization of some individual specimens from these sites showed fair stability. The mean directions of the sixteen sites, before and after cleaning, are plotted in figs. 1a and 1b respectively.

Typical a.f. demagnetization curves of the specimens from the significant sites are illustrated in fig. 2a and changes in directions on step-wise demagnetization are shown in fig. 2b. The viscous component of magnetization appears to have been removed at about 200 Oe. Specimen $3057A_2$ possesses higher coercivity of remanence than those indicated by specimens $3041A_1$ and $3095A_0$. The specimens from sites 11 and 12, excluding the surface specimens, were observed to be stable and to have high coercivity of remanence ($3057A_2$ type), whereas the specimens from other significant sites carried weak NRM and low coercivity of remanence ($3041A_1$ type).

It was suspected that alteration due to weathering of the surface samples was producing a stable secondary component, and giving rise to the observed scatter of directions within a site. To test this hypothesis and to find the depth to which this secondary component of magnetization is reflected in the various magnetic parameters, samples were collected from a vertical profile at an interval of about 30 cm to a maximum depth of 4.3 m at site 11.

The specimens from the vertical profile were treated in 300 Oe peak field. The variations in normalized intensity ($J(300)/J_0$), declinations and inclinations (D and I), and Curie points of the samples with depth are shown in fig. 3. The change in inclination with depth is gradual, $+22.5^\circ$ to -10.3° while the declination varies from 183.0° to 189.0° before showing a rapid change to 13° at the surface. The surface samples were further treated up to 500 Oe peak field. No further major changes in directions and inclinations were observed. The direction of the remanent magnetization thus became completely and stably reversed through about 180° at the surface. A typical demagnetization curve for the surface specimens is illustrated in fig. 2a (specimen $3095A_0$).

Samples of Nairobi phonolite were also collected to check whether similar variations in the magnetic properties with depth are repeated in other phonolites. The Nairobi phonolite is a different rock unit altogether and has been dated at 5.2 my [4]. The sampling was carried out at a quarry near Ruaraka, about 11.2 km from Nairobi. Fig. 4 shows the variations in magnetic properties of the Nairobi phonolite after "cleaning" the specimens in 400 Oe peak field. These variations in properties with depth are similar to those observed in the Kapiti phonolite except with regard to the Curie

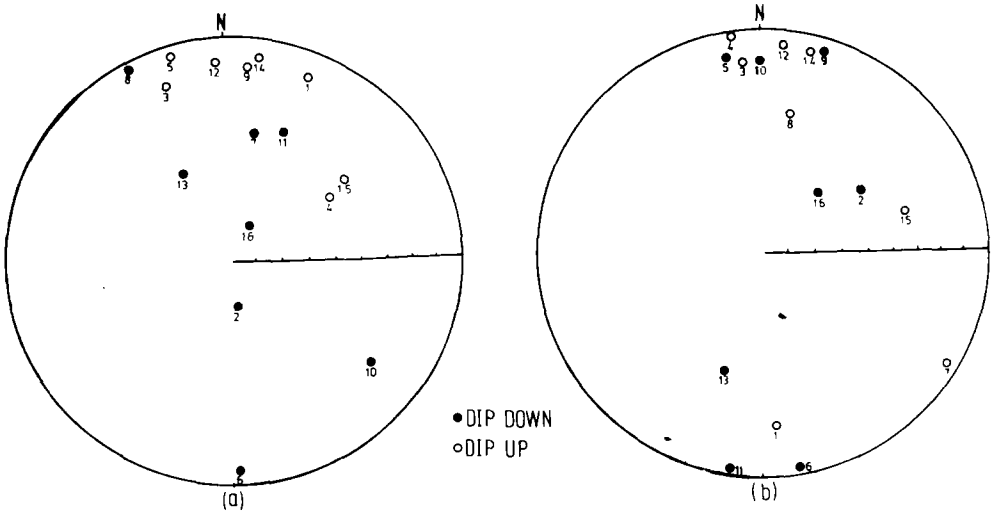


Fig. 1.

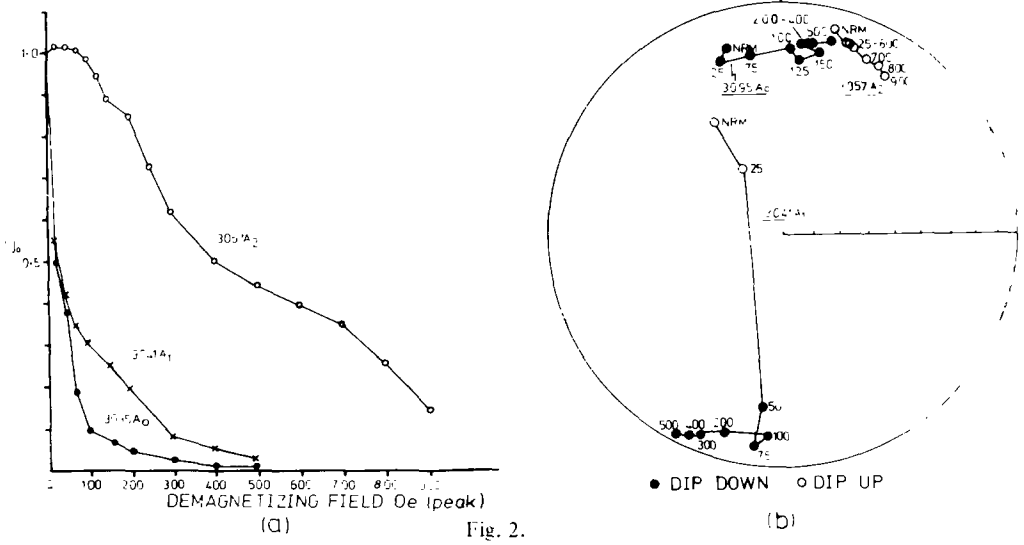


Fig. 2.

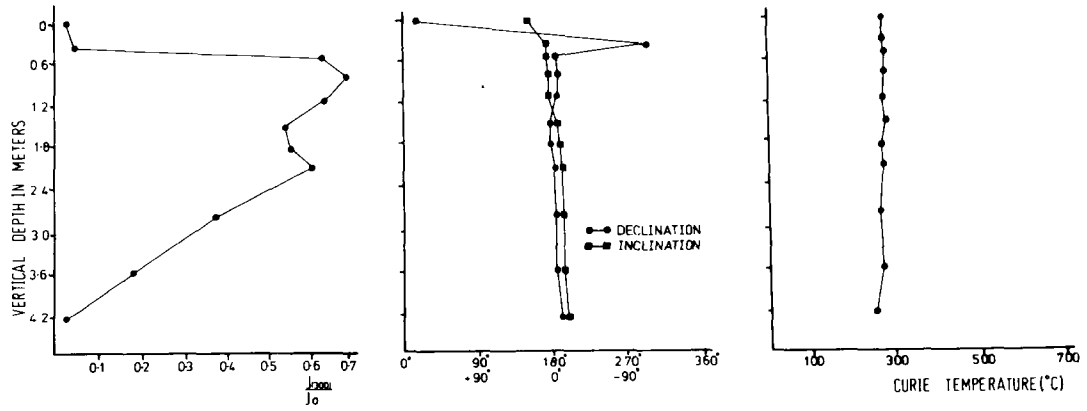


Fig. 3.

Table 1
Results

Site number	N	n	Mean D^{o}	Before cleaning Mean I^{o}	R	k	After cleaning Mean D^{o}	Mean I^{o}	R	k	α^{o}_{95}	Watson test	Pole position Lat. ^{o}N	Long. ^{o}L
1	8	16	35.1	−11.1	4.470	2.0	175.3	−22.6	6.448	4.5	29.8	non-random	−76.4	17.4
2	4	7	168.3	+72.1	2.226	1.6	59.5	+46.2	2.138	1.4	–	random	–	–
3	13	14	342.3	−19.8	7.021	2.0	355.4	−16.1	6.318	1.9	43.8	non-random	81.5	250.0
4	10	11	59.7	−45.5	1.038	1.0	353.4	− 1.1	5.412	2.0	–	random		–
5	8	8	346.3	− 5.7	5.161	2.5	351.4	+12.9	5.880	3.3	35.8	non-random	77.9	349.6
6	22	22	178.5	+ 5.4	11.712	2.0	170.9	+ 0.8	7.912	1.6	41.9	non-random	−80.9	309.8
7	4	4	13.0	+41.5	2.450	1.9	121.9	1.9	1.821	1.4	–	random	–	–
8	20	25	334.2	+ 3.1	4.905	1.3	12.1	−37.2	7.193	2.0	45.8	non-random	67.3	187.2
9	11	14	7.1	−14.1	3.438	1.4	18.7	+ 6.1	4.997	1.7	–	random	–	–
10	9	11	127.3	+22.5	2.279	1.3	0.5	+15.0	4.892	1.9	51.5	non-random	81.0	40.8
11	24	24	25.2	+38.3	12.023	1.9	188.2	+ 0.1	23.137	26.7	5.1	non-random	−81.8	121.9
12	20	20	358.6	−11.7	19.068	20.4	6.4	− 6.9	19.655	55.0	4.6	non-random	83.3	143.3
13	7	7	234.0	+54.0	4.337	2.3	197.4	+43.1	5.072	5.4	31.5	non-random	−60.4	183.9
14	4	4	10.1	− 9.6	3.931	43.5	14.0	− 6.8	3.960	75.5	20.3	non-random	75.8	133.1
15	8	8	56.9	−36.1	5.711	3.1	74.8	−33.7	5.114	3.2	40.6	non-random	14.8	146.1
16	4	4	33.9	+75.6	3.991	3.3	46.3	+60.2	3.773	13.2	24.6	non-random	29.7	76.9

N = Number of samples.
n = Number of specimens.
D = Declination in degrees east of true north.
I = Inclination in degrees measured positive downward.
R = Resultant of N vectors.
k = Fischer's precision parameter.
α^{o}_{95} = Radius of circle of 95% confidence.

Table 2

Pole from mean direction of significant sites									Mean of poles of significant sites				
					Palaeomagnetic North Pole							Palaeomagnetic North Pole	
N	Mean D^{o}	Mean I^{o}	k	α^{o}_{95}	Lat. ^{o}N	Long. ^{o}E	δ_p	δ_m	N	K	A^{o}_{95}	Lat. ^{o}N	Long. ^{o}E
12	9.2	4.2	5.3	20.9	79.3	128.3	8.4	21.6	12	7.5	16.9	80.8	117.7

N = Number of sites.
D, I = Declination and inclination as in table 1.
k, K = Fisher's precision parameter estimate.
$\alpha^{o}_{95}, A^{o}_{95}$ = Radius of circle of 95% confidence.
δ_p, δ_m = Semiaxes of 95% confidence oval.

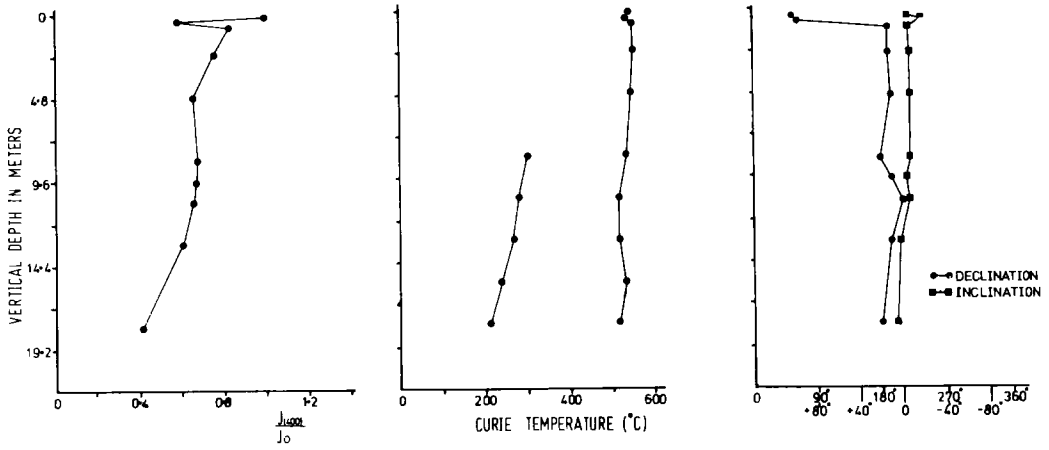


Fig. 4.

point. The Nairobi phonolite specimens possess two Curie points below a depth of about 8 m. The single low Curie point of the Kapiti phonolite was rather poorly defined due to the long tails of the heating curves. Similar behaviour was also observed by Ade Hall and others [10] and Raja [11]. The direction of remanent magnetization once again changed totally at the surface.

These magnetic properties which vary with depth may be partly of secondary origin. The surface samples possess a hard secondary component which could not be removed by a 500 Oe field. At higher fields the magnetization became too weak for measurement. Thin section petrology reveals that the surface samples are highly oxidized with the production of aggregates of magnetite dust from the alterations of iron bearing silicates [12]. This secondary component may be a chemical remanent magnetization acquired in the ambient field at the time of chemical transition. Although the magnetic properties vary considerably with depth, a reasonable directional grouping is observed five centimeters below the surface.

5. Upper Miocene pole

The mean Upper Miocene pole position is calculated at 81°N, 118°E (A_{95} = 17°) using the twelve pole positions of the significant sites of the Kapiti phonolite (table 2). The directions and the Pole positions of sites 1, 6, 11 and 13 were reversed through 180° to facilitate the calculations of the mean Pole position.

Miocene pole positions for Africa have been quoted by various authors. Watkins and others [13] give a pole 85°N, 136½°E (A_{95} = 3°), Raja [11] gives one at 85°N, 163½°E (A_{95} = 2.5°) (North Turkana lavas, Kenya), and Reilly [14] quoted one at 85°N, 140°E (A_{95} = 3.5°) for the Rift Valley Volcanics. The pole position of the Kapiti phonolite is not significantly different from the above Miocene poles and also from the present pole.

6. Conclusion

The variations in the magnetic parameters with depth shown by the results of the two vertical profiles are of great importance in Palaeomagnetic studies of tropical rocks which have few artificial exposures. The stable secondary magnetization observed in the surface samples, may be a hard secondary remanent magnetization acquired in the earth's ambient field and the local field within a rock. This secondary magnetization may be responsible for the scattered directions observed within a site. For phonolites, magnetic directions should be evaluated from samples obtained at a minimum depth of about 5 cm from the surface.

The existence of mixed polarities in the Kapiti phonolite is confirmed. All sites of Fort Hall area, north of Doinyo Sapuk (Map 1) have reversed polarity. This is in keeping with the mixed polarities observed in Kisumu and Kericho phonolites (12.3 ± 0.2 my and

12.4 ± 0.2 my respectively) of the Western flank of the Gregory Rift Valley [11]. The Kapiti phonolite has been observed to be about 150 m thick at some localities where it is composed of several flows laminated by argillaceous sediments [15]. The occurrence of normal, reverse and intermediate polarities suggests that extrusion of lava was intermittent over a period when the magnetic field of the earth underwent quick transitions [16].

The pole position for the Upper Miocene is calculated at 81°N, 118°E (A_{95} = 17°) and is in fair agreement with other Miocene poles of Africa.

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